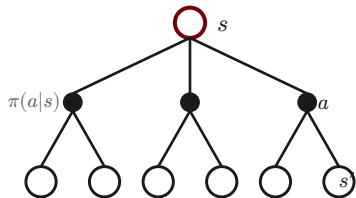
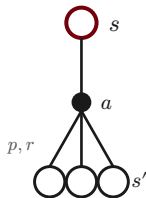


$v_\pi$  backup



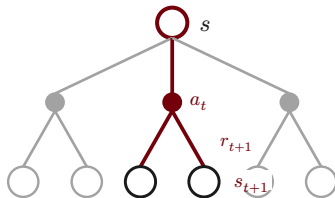
average over **both** actions ( $\pi$ )  
and outcomes ( $p$ )

$q_\pi$  backup



action  $a$  **fixed**; average  
over outcomes ( $p$ ) only

sample (TD) backup



follow **one** sampled  
transition — no averaging

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r | s,a) [r + \gamma v_\pi(s')]$$

the backup diagram is this expectation drawn as a tree: outer sum = action fan, inner sum = outcome fan, leaves =  $v_{\pi(s')}$ .

○ state node (open circle)

● action node (filled circle)

○ node being updated

↑ backup direction